

TrES-2b

R-0058

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-2_210902 · created 2026-07-15T13:50:49+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459459.69172 +/- 0.00084 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.147 +/- 0.044
Transit depth (Rp/Rs)^2	2.17 +/- 1.29 [%]
Orbital Inclination (inc)	83.93 +/- 2.89
Airmass coefficient 1 (a1)	0.182 +/- 0.061
Airmass coefficient 2 (a2)	1.72 +/- 1.83
Scatter in the residuals of the lightcurve fit is	40.05 %
Best Comparison Star	#3 - [81, 128]
Optimal Method	PSF photometry
Transit Duration (day)	0.067 +/- 0.041

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.147 ± 0.044 vs 0.1254 (deviation 0.39σ)
Duration fit vs published	0.067 d vs 0.0746 d (deviation 0.15σ)
Mid-time O–C	0.3 min
Depth SNR	2.6
Residual scatter	40.05 %

Light curve

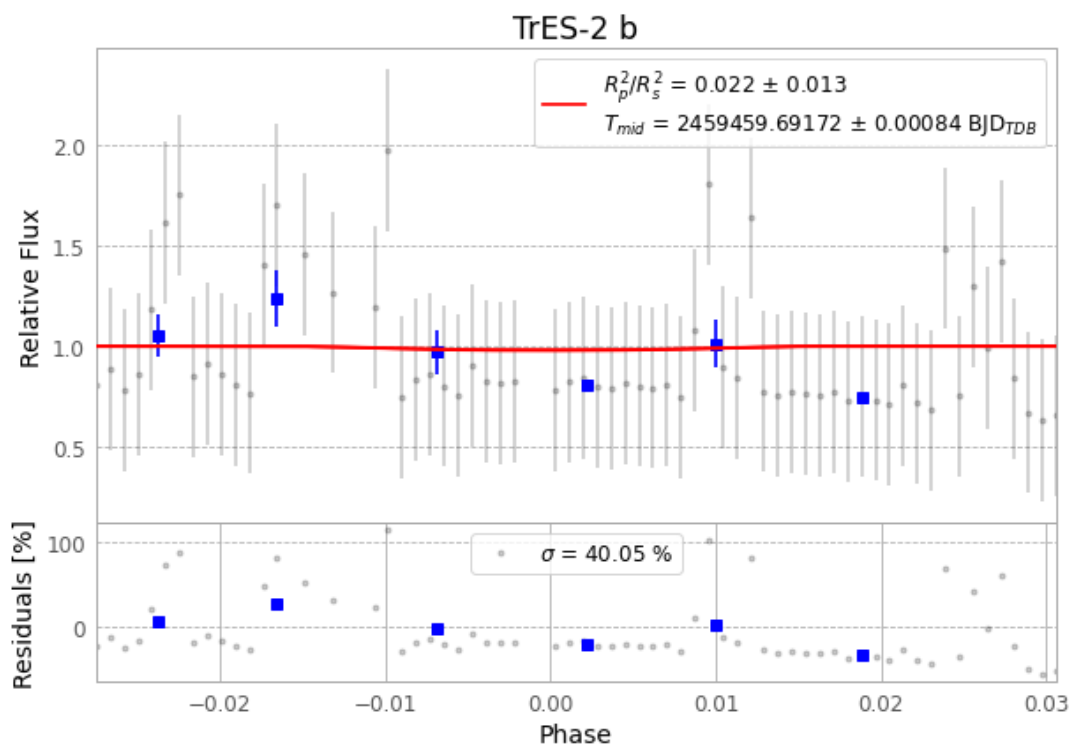


Figure 1 — FinalLightCurve_TrES-2 b_2021-09-01.png (SHA-256 f12c0cd412818ce7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [403, 214], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4179a70ab83471a3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

66 FITS frames (43 MB), TRES-2210902025412.FITS → TRES-2210902062112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-210902.log (e6296826acf4...), FinalLightCurve_TrES-2 b_2021-09-01.png (f12c0cd41281...), FinalLightCurve_TrES-2 b_2021-09-01.csv (4e1747cf72d3...)

Compute resources

Wall time 222.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 16:40Z.